

Data Analysis with Supervised Machine Learning

Objective

The objective of this project is to develop a machine learning model that can perform a data analysis on synthetically generated data. You will be required to use one or more common supervised machine learning methods to perform the analysis.

Data

The data that will be used for this project will be synthetically generated. The data will consist of various features that may or may not be relevant to the target variable. The Python code below is responsible for generating the data you are supposed to work with.

Python

```
import numpy as np

def generate_synthetic_data(num_samples, num_features):

    #Generate random data with normal distribution

    X=np.random.normal(size=(num_samples, num_features))

    #Generate target variable y with a linear combination of a subset of features plus
    noise

    relevant_features = np.random.choice(num_features, size=int(num_features/2),
    replace=False)

    w=np.zeros(num_features)

    w[relevant_features] = np.random.normal(size=(len(relevant_features),)) noise =
    np.random.normal(scale=0.1, size=(num_samples,)) y =X[:,
    relevant_features].dot(w[relevant_features]) + noise

    #Introduce missing values in the data

    missing = np.random.choice(num_samples * num_features, size=int(num_samples *
    num_features * 0.1), replace=False) X.flat[missing] = np.nan return X, y

num_samples =1000

num_features = 5

X, y = generate_synthetic_data(num_samples, num_features)
```

Data Processing

You are required to perform data cleaning and dimensionality reduction on the data set. Data cleaning will involve removing any missing or irrelevant data, and converting the data into a format that can be used by the machine learning algorithm. Dimensionality reduction will be performed using PCA (Principal Component Analysis), which is a common technique for reducing the number of features in a data set while retaining the most important information.

Methodology

You have to apply supervised machine learning methods on the data set at hand. This may include regression, decision trees, support vector machines, or other algorithms. Evaluate the performance of the model you used by comparing the predictions made by the model to the actual data.

Deliverable

1. The final deliverable of this project will be a report that includes the following:
A description of the data set and the preprocessing steps performed
2. An explanation of the machine learning method(s) used, including any choices made regarding parameters or hyperparameters.
3. Evaluation of the model's performance using relevant metrics
4. A discussion of the results and any limitations of the model

The report should be written in a clear and concise manner and should include appropriate visualizations and tables to support the analysis. Additionally, the code used to perform the analysis should be included to the repository as well.